Part 3 — Functional Dependency Inference

1. Given Functional Dependencies

The following functional dependencies (FDs) are given:

- **FD_1:** EmpID → Dept
- **FD_2:** Title → Grade
- **FD_3:** Dept, Grade → Bonus

These dependencies describe the relationships between employee IDs, departments, job titles, grades, and bonuses.

2. Computing the Closure of Q = {Dept, Title}

The initial set of attributes is:

Q = {Dept, Title}

To compute the closure Q^+, we apply the given functional dependencies step by step.

Step 1: Initially, Dept and Title are included in Q^+.

Q^+ = {Dept, Title}

Step 2: Using FD_2 (Title → Grade), we can infer Grade because Title is already in Q^+.

Q^+ = {Dept, Title, Grade}

Step 3: Using FD_3 (Dept, Grade → Bonus), we can infer Bonus because both Dept and Grade are now in Q^+.

Q^+ = {Dept, Title, Grade, Bonus}

Step 4: FD_1 (EmpID → Dept) cannot be used to infer EmpID because the dependency determines Dept from EmpID, not the other way around.

Therefore, the final closure is:

Q^+ = {Dept, Title, Grade, Bonus}

3. Proving That Bonus ∈ Q^+

Since Bonus is included in the computed closure Q^+, we conclude that:

Bonus ∈ Q^+

This means that the bonus can be inferred from Dept and Title through the following chain:

Title → Grade

Dept, Grade → Bonus

Consequently, a user who knows an employee's department and job title may be able to infer the corresponding bonus, even if the bonus column is not directly accessible.

This represents a potential inference risk when the functional dependencies accurately describe the underlying data.

4. Security Decision: Reject or Transform

The original query requesting Dept and Title should be evaluated because these attributes can indirectly reveal Bonus through the given functional dependencies.

For this implementation, the chosen approach is **Transform** rather than completely rejecting all access to employee information.

The security design applies the following controls:

- Direct access to the sensitive Employee table is denied to the student role.
- Access to the TitleGrade and BonusMatrix tables is denied to reduce the availability of information that could be used for inference.
- The view v_employee_safe exposes only EmpID and Dept.
- The view v_safe exposes department-level employee counts without revealing individual bonus values.

These controls limit direct access to sensitive information and provide safer alternatives for permitted queries.

5. Conclusion

The closure calculation demonstrates that Bonus belongs to the closure of {Dept, Title}. Therefore, the bonus can be inferred using the specified functional dependencies.

The SQL implementation combines access-control permissions with restricted views to reduce the risk of exposing sensitive employee information. However, these controls should be tested under the student user's permissions, and the views should be evaluated for possible inference through additional queries.